GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Challenge Name: Harnessing the Generative Quantum Eigensolver for Next-Generation Materials Design

Phase #: 3

Team Name: Entangled Trio

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1 Focus Area, Rationale, and Stakeholder Relevance

This project implements a dual extreme ultraviolet (EUV) simulation pipeline to model both the photoabsorption and subsequent core-level Auger electron decay in photoresist materials to evaluate their chemical sensitivity and performance. During EUV lithography, 92 eV exposure first induces photoabsorption to excite or ionize core electrons (e.g., Iodine 4d shell), which subsequently undergo Auger decay cascades. These secondary electrons initiate chemical reactions such as photoacid generation which create patterns in the resists. Understanding both the initial absorption and the subsequent electron emission is essential for optimizing resist sensitivity and resolution at sub-10 nm nodes. The interactive simulation code for our EUV simulation pipeline has been published as a pre-rendered Jupyter notebook [1].

The primary stakeholders and beneficiaries of this work include:

- **Semiconductor Manufacturers and Lithographers:** Guides the predictive design of photoresists to maximize light-absorption and secondary electron yield.
- **Materials Scientists:** Provides a robust computational framework for predicting electron cascade kinetics in halogenated compounds.

2 Technical Methodology and Hybrid Quantum Integration

Our EUV simulation pipeline is written in PennyLane [2], and models both photoabsorption and Auger decay:

1. **Photoabsorption Spectrum:** Using a time-domain Green's function propagated via a second-order Trotter product formula on a Compressed Double-Factorized (CDF) Hamiltonian, based on [3].
2. **Auger Electron KE Spectrum:** Using a Generative Quantum Eigensolver (GQE) [4] combined with the Quantum Self-Consistent Equation-of-Motion (q-sc-EOM) and One-Center Approximation (OCA), based on [5].

2.1 EUV Photoabsorption Spectrum Simulation

The EUV photoabsorption channel simulates the dipole-allowed valence-excitation spectrum under 92 eV radiation. We compute the transition dipole moment operators $\hat{m}_\rho$ in the molecular active space and project the ground state wavefunction to obtain initial states $\hat{m}_\rho|I\rangle$ along the Cartesian x, y, z coordinates. We then propagate the states in the time-domain after which measuring the time-dependent expectation values $\langle \hat{m}_\rho(t) \rangle$ yields the time-domain Green's function $G(t)$, which is Fourier-transformed to resolve the absorption spectrum in the frequency domain.

2.2 EUV Auger Emission: The GQE Challenge and Decay Modeling

Simulating the Auger emission spectrum requires modeling multi-electron electronic decay from core-ionized and doubly-ionized states. Since the decay transitions are highly sensitive to the ground-state reference, preparing a high-fidelity reference state represents a major GQE challenge. We address this using a hybrid quantum workflow consisting of three stages.

2.2.1 GQE Ground-State Reference Preparation

To prepare high-fidelity reference ground states, we employ a GQE that generates shallow quantum circuits. We implement two transformer architectures in PyTorch:

1. **GPTQE:** An autoregressive causal model based on nanoGPT that predicts discrete gate sequences.
2. **DiTQE:** A Diffusion Transformer-style architecture where the target ground-state energy acts as a continuous condition. DiTQE uses multi-headed self-attention with adaptive RMSNorm to modulate token embeddings, capturing global circuit topology and eliminating directional biases.

2.2.2 Excited-State Extraction via q-sc-EOM

Using the optimized GQE reference state $|0\rangle = U_{\mathrm{GQE}}|\mathrm{HF}\rangle$ as the vacuum, we extract excited states via the q-sc-EOM solver [6] across two channels:

- **Ionization Potential (IP) Space:** Resolves core-ionized ($N-1$ electrons) doublet states $|\mathrm{IP}_I\rangle$.
- **Double Ionization Potential (DIP) Space:** Resolves doubly-ionized ($N-2$ electrons) singlet states $|\mathrm{DIP}_K\rangle$.

We compute the transition reduced density matrix (RDM) $R_{KI;\alpha\beta} = \langle \mathrm{DIP}_K|\hat{a}_c^\dagger \hat{a}_\beta \hat{a}_\alpha|\mathrm{IP}_I\rangle$ involving one-electron creation and two-electron annihilation.

2.2.3 Minimal Basis Projection and One-Center Approximation (OCA)

To computationally simulate the Auger decay of H_2O and photoresist targets:

- **MBS Localization:** MO coefficients C in the primitive CGTO basis are localized onto the core-hole emitter site via the projection $D = T^{-1}UC$. The OCA restricts the final two-electron Coulomb elements strictly to this single atomic center, neglecting multi-center continuum wave distributions.
- **Three-Body Transition RDM:** Evaluates localized decay amplitudes via $B_{KI;lm} = \sum_{\alpha,\beta} R_{KI;\alpha\beta} V_{lm\alpha\beta}$.
- **Spectral Resolution:** Computes partial decay rates via Fermi's Golden Rule ($\Gamma_{KI;lm} = 2\pi|B_{KI;lm}|^2$), determines kinetic energies $E_{\mathrm{kin}} = E_{\mathrm{IP}}^{(I)} - E_{\mathrm{DIP}}^{(K)}$, and we resolve the continuous spectrum via Lorentzian broadening at 1.5 eV.

2.3 Phase 3 Pipeline Enhancements: Scaling, GPU Acceleration, and Memory Optimization

In Phase 3, we introduced major architectural and computational enhancements to scale the simulation pipeline from toy molecules (H_2O) to realistic halogenated photoresist materials:

1. **Target Molecule Scaling to IMePh (C_7H_7IO):** We integrated automated 3D geometry querying via PubChem [7] for 4-iodo-2-methylphenol (IMePh), a key EUV photoresist target containing heavy Iodine.
2. **Effective Core Potential (ECP) Integration:** To model heavy Iodine without prohibitive qubit counts, we incorporated a `def2-SVP` small-core ECP that replaces 46 non-participating core electrons while explicitly maintaining the EUV-active 4d shell. Combined with Active Space Selection (AVAS), this reduces the required active space to CAS(16e, 14o) / 28 qubits for single-GPU execution or CAS(24e, 18o) / 36 qubits for cluster execution.
3. **Memory-Efficient Wavefunction Dipole Projection:** We replaced $2^N \times 2^N$ sparse matrix Kronecker products for dipole projections with PySCF's 1-body contraction (`fci.direct_spin1.contract_1e`). Statevector representations are stored as compact sparse dictionaries (~150 KB) and generated on-demand per Cartesian coordinate,

reducing peak host RAM consumption from > 68 GB down to < 1.5 GB.

4. **GPU Acceleration and Precision Optimization:** We configured PennyLane QNodes to execute on `lightning.gpu` using cuQuantum/cuStateVec backends. By configuring single-precision `complex64` statevector representations across all Trotter evolution and q-sc-EOM emission steps, PCIe data transfer overheads were halved and GPU memory bandwidth throughput was doubled.
5. **CDF Fragment Truncation:** We implemented configurable double-factorization fragment truncation (`MAX_CDF_FRAGMENTS = 10`), retaining dominant two-body tensor fragments to reduce Trotter gate counts per step by $\sim 70\%$ without loss of physical accuracy.
6. **Single-Pass Quantum Evaluation without Snapshots:** For GQE candidate evaluation, we replaced multi-tape `qml.snapshots` with a direct single-pass `final_energy_circuit` QNode, reducing quantum circuit tape executions per evaluation check by $5\times$.

3 Data Modeling and Benchmarking Strategy

Our data modeling strategy utilizes the standard STO-3G basis set. To scale both the photoabsorption and the core-level emission calculations, we incorporate four acceleration techniques:

1. **Active Space Selection (AVAS):** Trims molecular orbitals down to active valence electrons and orbitals (e.g., CAS(16e, 14o) for 28 qubits or CAS(16e, 16o) for 32 qubits in IMePh) to maintain classical tractability.
2. **Effective Core Potential (ECP):** Replaces non-participating core electrons of heavy halogen atoms with pseudopotentials while maintaining core 4d excited state channels.
3. **Compressed Double Factorization (CDF):** Compresses the molecular Hamiltonian terms using a Block-Invariant Symmetry Shift (BLISS) and L2-regularized numerical fitting, minimizing the one-norm of the two-body operator.
4. **Automated PubChem 3D Integration:** Fetches exact 3D molecular structures for photoresist target candidates directly from PubChem databases.

For validation, our classical benchmarks include exact FCI from PySCF [8] to validate GQE training, and RASSI in OpenMolcas [9] to validate the Auger spectrum simulation.

4 Experimental Results and Discussion

For the **absorption** simulation, spectrum in the 10–100 eV range shows good agreement with the exact classical CASCI baseline, resolving the physical valence excitations in the 10–45 eV regime and dropping to a flat zero baseline in the 45–100 eV range. This confirms that H_2O has no dipole-allowed transitions near the 92 eV EUV regime in a minimal STO-3G basis, which is expected. This serves as a key stepping stone for photoresist-adjacent molecules, e.g., IMePh (4-iodo-2-methylphenol, C_7H_7IO) where Iodine 4d core-excitations are active at 92 eV.

For the **emission** simulation, GQE reference state preparation is the core challenge. The GQE training was evaluated over 10,000 epochs. GQE energy statistics compared against random sequences are summarized in Table 1.

The GPT model successfully optimized the excitation sequences, lowering the minimum energy to −74.9650 Hartrees. DiTQE was used to evaluate the benefits of energy-conditioned generative transformer architectures against the GPTQE baseline. We identified that the initial poor performance of the DiTQE model relative to the baseline was caused by a causal leakage bug in the multi-headed self-attention (MHA) block. Because the attention mechanism did not originally apply a lower-triangular causal mask, the model was able to look ahead at future tokens during

Table 1: GQE Ground State Energy (Hartrees) of H_2O

Source	Average Energy	Minimum Energy	Maximum Energy	Error to Ground State
GPT	-74.9533	-74.9650	-74.9131	0.0475
DiT	-74.9327	-74.9715	-74.7391	0.0409
FCI	-75.0125	–	–	0.0000

autoregressive training. Correcting this by enforcing causal masking in the MHA block resolved this look-ahead data leak, aligning training attention with autoregressive sequential generation and enabling DiTQE to learn sequence representations without cheating.

Using GQE reference states, q-sc-EOM resolved the excited states, while MBS and OCA yielded Auger decay rates. We successfully reproduce the main peak at E_{kin} = 502.4 eV, which aligns with [5] and experimental results [10].

5 Quantum Platform Justification and Resource Needs

Requested: 8xB200-180GB-SXM6 (Inferior alternative: 8xH100-80GB-SXM5).
FP64 state-vector memory: 35 qubits (549.7 GB), 36 qubits (1,099.5 GB), 37 qubits (2,199.0 GB).
Our rationale for selecting Blackwell (B200) over H100:

- **Latency**: Halves the GPU count for NVLink distribution vs H100 (2 vs 4 GPUs for 34 qubits); SXM6 NVLink-5 bandwidth (1.8 TB/s) minimizes inter-GPU communication bottleneck.
- **Capacity**: 8xB200 offers 1,440 GB VRAM, fitting the 36-qubit space (1,099.5 GB) on a single node. 8xH100 is capped at 640 GB and will go out of memory.
- **Precision**: SXM6 native FP64 throughput prevents rounding errors in the kinetic energy distributions.

6 Gap Analysis and Implementation Roadmap

In Phase 3, we successfully resolved major computational bottlenecks by integrating Effective Core Potentials (ECP), AVAS active space selection, PySCF 1-body wavefunction projection, cuQuantum GPU statevector acceleration, single-precision `complex64` execution, and single-pass GQE evaluation. These advancements allowed us to scale the pipeline to 28–32 qubits for large halogenated photoresists like IMePh (C_7H_7IO). Our final roadmap targets 36-qubit scaling on multi-node GPU clusters (8xB200), full double-factorization BLISS optimization, and multi-center continuum wave modeling for Auger decay.

References

[1] N. Venkat. Quantum Chemistry for Spectroscopy in EUV Lithography. `nirajvenkat.com/mb_nb`, 2026.

[2] V. Bergholm et al. PennyLane: Automatic Differentiation of Hybrid Quantum-Classical Computations. *arXiv preprint arXiv:1811.04968*, 2018.

[3] A. Kharazi et al. Quantum Simulations for Extreme Ultraviolet Photolithography. *arXiv preprint arXiv:2602.20234*, 2026.

[4] K. Nakaji et al. Generative Quantum Eigensolver for Excitation Spectra. *Physical Review Research*, 6(1):013142, 2024.

[5] L. Keithley et al. Auger Spectroscopy via Generative Quantum Eigensolver: A Quantum Approach to Molecular Excitations. *arXiv preprint arXiv:2603.12859v1*, 2026.

[6] A. Asthana et al. Quantum self-consistent equation-of-motion method for computing molecular excitation energies, ionization potentials, and electron affinities on a quantum computer. *Chem. Sci.*, 14:2405–2418, 2023.

[7] S. Kim et al. PubChem in 2021: new data, new tools and new services. *Nucleic Acids Research*, 49(D1):D1388–D1395, 2021.

[8] Q. Sun et al. Recent Developments in the PySCF Program Package. *The Journal of Chemical Physics*, 153(2):024109, 2020.

[9] I. F. Galvan et al. OpenMolcas: From the Ground State to Excited States. *Journal of Chemical Theory and Computation*, 15(11):5925–5940, 2019.

[10] W. E. Moddeman and et. al. 1973.